Notes: Giglio, Maggiori, Stroebel, & Utkus (AER, 2021)

Five Facts about Beliefs and Portfolios

Contents

1 Big picture

- **Question.** Are survey-elicited beliefs economically meaningful for household portfolio choice, trading, and macro-finance modeling?
- **Setting.** The paper studies the GMSU-Vanguard survey, a new online expectations survey administered every two months to US Vanguard clients.
- **Data advantage.** The survey elicits quantitative beliefs about stock returns, GDP growth, bond returns, return distributions, confidence, and question difficulty, then links those beliefs to administrative data on portfolio holdings, trading, and login behavior.
- **Sample.** The first 21 survey waves cover February 2017 to June 2020 and generate 46,419 completed responses. The respondents are financially relevant: the average respondent holds more than half a million dollars at Vanguard.
- **Fact 1.** Beliefs are reflected in portfolios. A 1 percentage point higher expected 1-year stock return is associated with about a 0.7 percentage point higher equity share. This is statistically strong but far smaller than a frictionless Merton benchmark.
- **Heterogeneity in Fact 1.** Belief-to-portfolio pass-through is stronger for wealthier investors, tax-advantaged retail accounts, investors who trade more often, investors who log in more frequently, and investors who are more confident in their beliefs.
- **Fact 2.** Belief changes do not strongly predict when people trade. Conditional on trading, however, beliefs help explain both trade direction and trade magnitude.
- **Fact 3.** Beliefs are dominated by persistent individual heterogeneity. Some investors are systematically optimistic and others systematically pessimistic. Time variation in average beliefs is much smaller than cross-sectional disagreement.
- **Fact 4.** Expected cash-flow growth and expected stock returns are positively related, both across investors and within investors over time.
- **Fact 5.** Investors who assign higher probability to stock-market disasters expect lower stock returns, both across investors and within investors.
- **Economic takeaway.** Survey beliefs are useful, but macro-finance models should not treat them as representative-agent sufficient statistics. The important objects are heterogeneous, persistent beliefs, weak average portfolio pass-through, and infrequent trading.

2 Data and measurement

2.1 Why the GMSU-Vanguard setting is useful

- The survey is administered by Vanguard to a large sample of clients who actually own risky and safe assets.
- Respondents are not hypothetical investors. Their survey answers can be linked to their real portfolio shares, transaction records, and attention measures.
- The questions are quantitative, making the answers usable for macro-finance calibration.
- The repeated survey waves create a panel, allowing the authors to distinguish time-series changes from persistent investor fixed effects.

Interpretation: The setting addresses common objections to survey expectations: small samples, qualitative answers, and inability to connect stated beliefs to actual behavior.

2.2 Survey design

The survey elicits beliefs on three blocks:

- **Expected stock market returns:** expected annualized US stock return over 1 year and 10 years, plus probabilities that the 1-year return lies in five buckets: below -30%, $[-30\%, -10\%]$, $[-10\%, 30\%]$, $[30\%, 40\%]$, and above 40%.
- **Expected real GDP growth:** expected annualized growth over 3 years and 10 years, plus probabilities that 3-year annualized growth lies in five buckets: below -3%, $[-3\%, 0\%]$, $[0\%, 3\%]$, $[3\%, 9\%]$, and above 9%.
- **Expected bond returns:** expected 1-year return on a 10-year US government zero-coupon bond.

After each block, respondents report confidence and difficulty on five-point scales.

Interpretation: The survey asks for both point forecasts and subjective distributions. This lets the authors study expected returns, perceived variance, and rare-disaster probabilities separately.

2.3 Survey sample and response rates

- The first 21 waves run from February 2017 to June 2020.
- The initial invitation pool is mostly retail investors, with about 20% from employer-sponsored retirement accounts.
- First-contact response rates are around 2.5–4%.
- Re-response rates among prior respondents are much higher, around 10–15% in steady state.
- The survey generates more than 2,000 responses per wave on average.
- More than 40% of responses come from individuals who respond to at least four waves.

Interpretation: The sample is selected toward older, wealthier, and more active investors, but this is useful for asset-pricing questions because these investors own substantial wealth and can move portfolios.

2.4 Administrative portfolio data

For each respondent, Vanguard administrative data provide:

- total wealth at Vanguard;
- portfolio shares in equity, fixed income, cash, and other assets;
- number and type of assets held;
- account type, including retail and defined-contribution accounts;
- active trades and portfolio turnover;
- login frequency and time spent on the Vanguard website.

Interpretation: The key empirical advantage is not merely surveying beliefs. It is matching beliefs to high-stakes portfolio data measured outside the survey.

2.5 Main belief variables

The paper repeatedly uses the following belief objects:

- $E_{i,t}[R_{1y}]$: respondent i's expected 1-year stock return at wave t;
- $E_{i,t}[R_{10y}]$: expected 10-year annualized stock return;
- $Pr_{i,t}(R_{1y} < -30\%)$: subjective probability of a stock-market disaster;
- $E_{i,t}[G_{3y}]$: expected 3-year annualized GDP growth;
- $E_{i,t}[G_{10y}]$: expected 10-year annualized GDP growth;
- confidence and difficulty scores for each survey block.

Interpretation: The paper is especially interested in whether expected stock returns, expected cash-flow growth, and disaster probabilities move together in ways that asset-pricing models would require.

2.6 Main portfolio and trading variables

- **Equity share:** fraction of portfolio allocated to equities, after decomposing mutual funds using Morningstar holdings.
- **Fixed income share:** fraction in fixed income instruments.
- **Cash share:** fraction in cash and cash-equivalent holdings.
- **Trading indicator:** whether the investor actively changed equity share by at least one percentage point between two survey responses.
- **Trade direction:** whether the active trade increased the equity share.
- **Trade magnitude:** change in equity share conditional on trading.

Interpretation: The trading analysis separates the extensive margin, whether a person trades at all, from the intensive margin, how beliefs affect the trade after a person trades.

3 Models and empirical specifications

3.1 Frictionless portfolio benchmark

A standard Merton-style benchmark implies an equity share of the form

$$\omega_i = \frac{E_i[R] - r_f}{\gamma_i \sigma_i^2}, \tag{1}$$

where $E_i[R] - r_f$ is the investor's expected excess stock return, γ_i is risk aversion, and σ_i^2 is perceived return variance.

Interpretation: In a frictionless benchmark, the pass-through from expected returns to risky asset demand should be large. The paper's central empirical result is that the observed average pass-through is positive but much smaller.

3.2 Beliefs and equity shares

The baseline portfolio regression is

$$EquityShare_{i,t} = \alpha + \beta E_{i,t}[R_{1y}] + \Gamma' X_{i,t} + \psi_t + \varepsilon_{i,t}. \tag{2}$$

- $EquityShare_{i,t}$ is the respondent's portfolio equity share in percent.
- $E_{i,t}[R_{1y}]$ is expected 1-year stock return in percent.
- $X_{i,t}$ includes demographics and other controls.
- ψ_t are survey-wave fixed effects.

Interpretation: The coefficient β measures how much a portfolio equity share changes when an investor reports a higher expected stock return.

3.3 Measurement-error correction

Classical measurement error in reported beliefs can attenuate β. The paper uses repeated belief questions and obviously related instruments, including the two stock-return expectation horizons, to implement ORIV-style corrections.

Interpretation: The key conclusion is that measurement error is not the main reason the estimated pass-through is small. Even corrected estimates remain far below the frictionless benchmark for the average investor.

3.4 Heterogeneous pass-through

The paper estimates heterogeneous specifications of the form

$$EquityShare_{i,t} = \alpha + \sum_g \beta_g \left(E_{i,t}[R_{1y}] \times 1\{i \in g\}\right) + \Gamma' X_{i,t} + \psi_t + \varepsilon_{i,t}, \tag{3}$$

where g indexes groups by wealth, account type, trading frequency, attention, confidence, or model-idealized behavior.

Interpretation: This asks whether the average low pass-through masks a more responsive subset of investors.

3.5 Belief changes and trading

For the window w between two survey responses, the paper estimates

$$\Delta EquityShare_{i,w} = \alpha + \beta E_{i,w-}[R_{1y}] + \gamma \Delta E_{i,w}[R_{1y}] + \delta EquityShare_{i,w-} + \Phi' X_{i,w} + \epsilon_{i,w}. \tag{4}$$

The same right-hand-side variables are also used with dependent variables indicating whether a trade occurs and, conditional on trading, whether the trade increases the equity share.

Interpretation: The equation separates initial beliefs from belief changes. Fact 2 is that belief changes matter mostly conditional on an investor already trading.

3.6 Decomposing belief variation

For a belief variable $B_{i,t}$, the paper estimates three regressions:

$$B_{i,t} = \chi_t + \varepsilon_{1,i,t}, \tag{5}$$
$$B_{i,t} = \phi_i + \varepsilon_{2,i,t}, \tag{6}$$
$$B_{i,t} = \phi_i + \chi_t + \varepsilon_{3,i,t}. \tag{7}$$

- χ_t captures common time-series movements in beliefs.
- ϕ_i captures persistent individual optimism or pessimism.

Interpretation: Comparing R^2 values tells whether belief variation is mostly aggregate time-series movement or persistent cross-sectional heterogeneity.

3.7 Demographics and persistent beliefs

The paper then asks whether observable characteristics explain individual belief fixed effects:

$$\widehat{\phi}_i = \alpha + \Gamma' X_i + \eta_i. \tag{8}$$

Interpretation: Demographics and experiences have some statistically significant relationships with beliefs, but their joint explanatory power is small.

3.8 Expected returns and expected cash-flow growth

The paper interprets Fact 4 through the Campbell-Shiller logic:

$$pd_t \approx E_{i,t} \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j} - E_{i,t} \sum_{j=0}^{\infty} \rho^j r_{t+1+j}. \tag{9}$$

Interpretation: Since current valuations are fixed for all investors at a point in time, disagreement about future cash-flow growth and disagreement about future returns cannot be modeled independently. The survey directly measures their joint distribution.

3.9 Rare disasters and expected returns

The rare-disaster regression is

$$E_{i,t}[R_{1y}] = \alpha + \beta Pr_{i,t}(R_{1y} < -30\%) + \gamma' X_{i,t} + \psi_t + \varepsilon_{i,t}. \tag{10}$$

Interpretation: The sign of β distinguishes different mechanisms. In the data, people who think disasters are more likely expect lower, not higher, stock returns.

4 Identification and empirical design

4.1 What makes the evidence credible

- Beliefs are directly elicited rather than inferred from prices.

- Portfolio holdings and trades come from administrative records rather than survey self-reports.
- The panel structure separates cross-sectional differences from within-person belief changes.
- Survey-wave fixed effects absorb common market conditions and aggregate belief shifts.
- Robustness checks address measurement error, account types, COVID-19, outliers, and alternative survey populations.

Interpretation: The paper is not a randomized experiment, but it is unusually strong descriptive evidence on the relationship between stated beliefs and actual financial behavior.

4.2 Why the trading margin matters

- Portfolio shares may respond weakly to beliefs because investors do not trade often.
- Beliefs may be meaningful but only become visible in portfolios when investors decide to transact.
- The paper finds little evidence that belief changes trigger trades, but conditional on trading, beliefs affect trade direction and size.

Interpretation: This supports models with infrequent trading and heterogeneous trading probabilities rather than frictionless continuous rebalancing.

4.3 Why persistent heterogeneity matters

- A representative-agent model using average survey beliefs misses most of the variation in the data.
- Persistent optimists and pessimists can trade against each other and redistribute wealth after realized shocks.
- Demographics explain only a small part of persistent heterogeneity.

Interpretation: The paper pushes macro-finance models toward heterogeneous-agent belief structures.

4.4 What the paper cannot fully prove

- Vanguard clients are not a random sample of all households.
- Response rates are low, so survey participation is selected.
- Portfolio holdings outside Vanguard are not fully observed.
- The data show relationships between beliefs and portfolios, not random assignment of beliefs.
- Survey answers may still contain nonclassical measurement error or strategic simplification.

Interpretation: The results are best read as powerful empirical moments for models, not as a complete causal theory of belief formation.

5 Tables and figures

The original visuals are directly cropped from the paper. Adjacent visuals are grouped where possible to save space and make comparison faster.

5.1 Figure 1 and Table 1: survey response and respondent characteristics

Read:

- Figure 1 shows stable first-contact response rates of roughly 2.5–4% and much higher re-response rates among prior respondents.
- The survey produces more than 2,000 completed responses per wave on average.
- Table 1 shows that respondents are older, wealthier, and more active than nonrespondents.
- The average respondent is 60.1 years old, 69% male, holds about $520,000 at Vanguard, and holds 67.5% of the portfolio in equity.

Economic message: The sample is selected, but selected in a useful direction for asset-pricing questions: respondents own substantial risky assets and make real portfolio choices.

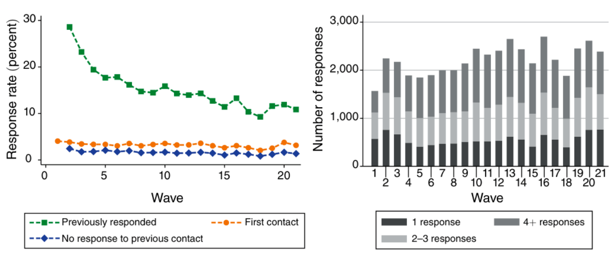

Figure 1: survey responses

TABLE 1—DEMOGRAPHICS: SURVEY RESPONDENTS AND NONRESPONDENTS

	Survey respondents				Non-respondents	
	Mean	P10	P50	P90	Mean	Difference
Age (years)	60.1	39.0	63.0	76.0	52.3	7.86
Male	0.69	0	1	1	0.54	0.16
Region						
Northeast	0.23	0	0	1	0.24	−0.01
Midwest	0.21	0	0	1	0.20	0.01
South	0.31	0	0	1	0.31	0.00
West	0.25	0	0	1	0.25	0.00
Total Vanguard wealth (k$)	520.0	23.2	227.8	1301.1	254.5	265.4
Length of Vanguard relationship (years)	17.11	5	17	29	14.32	2.78
Active trades/month	1.54	0.02	0.55	3.71	0.91	0.63
Monthly portfolio turnover (percent)	2.30	0.00	0.91	4.85	1.96	0.34
Days with log-ins/month	3.71	0.09	1.53	10.96	1.81	1.90
Total time spent/month (minutes)	31.0	1.1	11.9	72.5	12.3	18.72
Portfolio shares (percent)						
Equity	67.5	30.8	70.7	99.9	71.5	−4.02
Fixed income	20.9	0.0	15.8	48.4	17.3	3.54
Cash	10.1	0.0	1.7	31.4	10.1	−0.01
Other/unknown	1.6	0.0	0.0	4.2	1.1	0.48
Number of unique assets	7.55	1	4	17	4.65	2.90
Number of mutual funds	4.82	1	3	11	3.31	1.51
Number of ETFs	0.81	0	0	2	0.43	0.38
Number of stocks	1.80	0	0	4	0.87	0.92
Number of bonds	0.13	0	0	0	0.04	0.09

Notes: Table shows summary statistics on both the survey respondents and nonrespondents. Age, gender, location,

Table 1: respondent demographics

5.2 Table 2: survey responses and belief distributions

Read:

- The mean expected 1-year stock return is 4.64%, with a standard deviation of 6.08% across responses.
- The mean expected 10-year annualized stock return is 6.64%, with a smaller standard deviation of 3.85%.
- Respondents assign a mean probability of 5.3% to a 1-year stock return below −30%.
- The mean expected 3-year annualized GDP growth rate is 2.77%.
- Bond-return questions are perceived as more difficult than stock-return and GDP-growth questions.

Economic message: Investor beliefs are dispersed and quantitative. This dispersion is not a small nuisance; it is one of the paper's central empirical facts.

Table 2—Summary Statistics: Survey Responses

	Mean	SD	P10	P25	P50	P75	P90
Expected stock returns							
Expected 1Y stock return (percent)	4.64	6.08	−1	3	5	8	10
Expected 10Y stock return (percent p.a.)	6.64	3.85	3	5	6	8	10
Probability 1Y stock return in bucket (percent)							
Less than −30 percent	5.3	8.5	0	0	3	8	10
−30 percent to −10 percent	14.5	14.1	0	5	10	20	30
−10 percent to 30 percent	70.1	23.0	40	60	75	90	100
30 percent to 40 percent	7.3	10.6	0	0	5	10	20
More than 40 percent	2.7	6.3	0	0	0	5	10
Expected GDP growth							
Expected 3Y GDP growth (percent p.a.)	2.77	2.16	1	2	3	3	4
Expected 10Y GDP growth (percent p.a.)	3.15	2.74	2	2	3	3	5
Probability p.a. 3Y GDP growth in bucket (percent)							
Less than −3 percent	4.9	8.5	0	0	2	5	10
−3 percent to 0 percent	13.3	13.0	0	5	10	20	30
0 percent to 3 percent	58.7	25.8	20	40	60	80	90
3 percent to 9 percent	20.0	21.9	0	5	10	25	50
More than 9 percent	3.1	8.5	0	0	0	5	10
Expected bond returns							
Expected 1Y return of 10Y zero coupon bond (percent)	1.74	2.84	−1	1	2	3	4
Difficulty (“*not at all difficult*,” …, “*very difficult*”)							
Expected stock returns	2.35	0.98	1	2	2	3	4
Expected GDP growth	2.46	0.98	1	2	2	3	4
Expected bond returns	2.84	1.00	1	2	3	3	4
Confidence (“*not at all confident*,” …, “*very confident*”)							
Expected stock returns	3.04	0.85	2	3	3	4	4
Expected GDP growth	2.98	0.85	2	3	3	3	4
Expected bond returns	2.62	0.86	2	2	3	3	4
Time of responding to survey (seconds)	507	429	235	302	407	567	810

Notes: Table shows summary statistics of the answers across responses to the first 21 waves of the GMSU-Vanguard survey. The possible answers for *Difficulty* are 1 = “Not at all difficult,” 2 = “Not very difficult,” 3 = “Somewhat

5.3 Table 3 and Figure 2: beliefs and portfolio equity shares

Read:

- Table 3 shows that a 1 percentage point increase in expected 1-year stock returns is associated with about a 0.67–0.69 percentage point higher equity share in baseline specifications.
- Restricting to moderate expected-return answers or using ORIV corrections raises the coefficient but does not make it close to frictionless benchmark sensitivities.
- Figure 2 visually shows a positive conditional binscatter relationship between expected 1-year stock returns and equity shares.
- The relationship is statistically clear, but economically modest relative to standard Merton demand.

Economic message: Survey beliefs are not noise: they map into portfolios. But the mapping is weak enough that models must include trading frictions, taxes, inattention, defaults, or other wedges.

Table 3—Expected Returns and Portfolio Equity Shares

	Equity share (percent)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Expected 1Y stock return (percent)	0.672	0.690	1.164	0.634	0.818	1.177	1.188	1.146
	(0.034)	(0.034)	(0.061)	(0.053)	(0.042)	(0.062)	(0.069)	(0.071)
Expected 1Y stock return (percent) × assets > $225k				0.114				
				(0.067)				
Expected 1Y stock return (percent) × above median time							–0.021	
							(0.056)	
Expected 1Y stock return (percent) × closest prior trade 2 weeks before								0.479
								(0.356)
Expected 1Y stock return (percent) × closest prior trade 1 week before								0.460
								(0.220)
Expected 1Y stock return (percent) × closest next trade 1 week after								0.384
								(0.245)
Expected 1Y stock return (percent) × closest next trade 2 weeks after								0.188
								(0.266)
Controls	N	Y	Y	Y	Y	Y	Y	Y
ORIV	N	N	N	N	N	Y	Y	Y
Sample			*E*(return) 0–15 percent		Feb 2017– Feb 2020			Retail accounts
Observations	44,595	44,565	39,296	44,565	39,859	44,235	44,235	41,142

Table 3: expected returns and equity shares

racteristics that we estimate the sensitivity for. Standard errors are clustered at the respondent

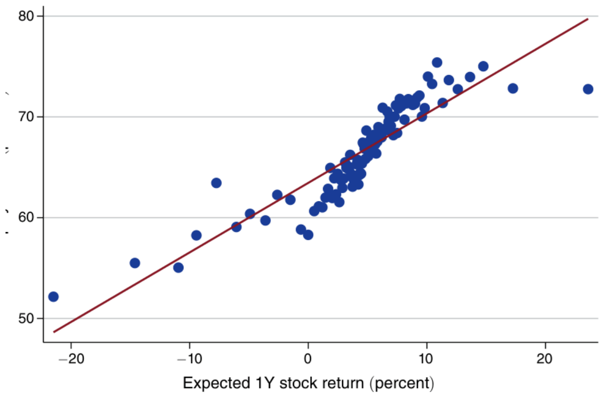

Figure 2: expected returns and equity shares

5.4 Table 4: heterogeneous belief-to-portfolio pass-through

Read:

- The sensitivity is larger in tax-advantaged accounts than in standard taxable retail accounts.
- It is stronger for investors with higher portfolio turnover.
- It is stronger for investors who log into Vanguard more frequently.
- It is stronger for investors who report higher confidence in their answers.
- The idealized group that is attentive, active, confident, and tax-advantaged has a coefficient around 3.55, several times the average coefficient.

Economic message: The average investor is not the only relevant object. Some investors behave closer to the frictionless benchmark, and these differences matter for equilibrium modeling.

5.5 Table 5: trading analysis

Read:

- Belief changes are related to changes in equity share over the window between survey responses.
- Belief changes have little explanatory power for the probability that a trade occurs.
- Conditional on trading, higher expected returns increase the probability of buying equity and increase the change in equity share.
- Initial equity shares matter negatively: investors with high starting equity shares tend to rebalance downward.

Economic message: Beliefs affect what investors do when they trade, but do not strongly explain when they trade. This is the evidence behind Fact 2.

5.6 Figure 3 and Tables 6–8: persistent belief heterogeneity

Read:

TABLE 4—EXPECTED RETURNS AND PORTFOLIOS: HETEROGENEITY

	Equity share (percent)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Expected 1Y stock return (percent)	1.198 (0.140)	1.419 (0.100)	1.270 (0.178)				
Expected 1Y stock return (percent) × monthly turnover $< 0.5\%$				0.737 (0.133)			
Expected 1Y stock return (percent) × monthly turnover $\in [0.5\%, 4\%]$				1.330 (0.084)			
Expected 1Y stock return (percent) × monthly turnover $> 4\%$				1.758 (0.231)			
Expected 1Y stock return (percent) × monthly Vanguard visits $\in (0,1)$					1.048 (0.116)		
Expected 1Y stock return (percent) × monthly Vanguard visits $\in [1,7)$					1.106 (0.072)		
Expected 1Y stock return (percent) × monthly Vanguard visits $\in [7,31)$					1.640 (0.135)		
Expected 1Y stock return (percent) × low confidence						0.810 (0.127)	
Expected 1Y stock return (percent) × medium confidence						1.229 (0.072)	
Expected 1Y stock return (percent) × high confidence						1.448 (0.268)	
Expected 1Y stock return (percent) × not idealized							1.338 (0.075)
Expected 1Y stock return (percent) × idealized							3.553 (1.329)
Controls + fixed effects	Y	Y	Y	Y	Y	Y	Y
Sample	Retail account	Retail account tax adv.	Defined contribution plans				Retail account tax adv.
Observations	19,921	19,921	4,112	41,252	44,235	44,235	34,486

Notes: Table shows results from regression (1), estimated using ORIV. The dependent variable is the equity share. In column 1 it is the equity share in individually managed retail accounts, in column 2 it is the equity share in individually managed tax-advantaged retail accounts, and in column 3 it is the equity share in institutionally managed retirement plans (defined contribution plans). In columns 5 and 6, it is pooled across the three types of accounts, while column 4 uses data from both types of retail accounts only. The sample in columns 1 and 2 is restricted to respondents holding both types of retail accounts. In column 6, *low confidence* corresponds to individuals who reported being "not at all confident" or "not very confident" in their answers about expected stock returns; *medium confidence* corresponds to individuals who report being "somewhat confident" or "very confident" about their answers; and *high confidence* corresponds to individuals who report being "extremely confident." *Idealized* respondents in column 7 are those whose behavior most closely corresponds to that of the assumptions in the frictionless model: they have average monthly portfolio turnover of at least 4 percent, they have at least seven logins a month,

TABLE 5—TRADING ANALYSIS

	Δ Equity share (pp)	Probability trade	Probability trade	Probability buy	Δ Equity share (pp)
	(1)	(2)	(3)	(4)	(5)
Δ expected 1Y stock return (pp)	0.229 (0.033)			0.977 (0.201)	0.587 (0.085)
Expected 1Y stock return (pp)	0.130 (0.019)	0.006 (0.130)		1.489 (0.199)	0.395 (0.062)
Lagged equity share (pp)	−0.048 (0.004)	−0.121 (0.021)		−0.337 (0.031)	−0.161 (0.011)
|Δ expected 1Y stock return (pp)|		0.258 (0.223)			
Extreme equity share dummies	Y	Y	Y	Y	Y
Time between wave dummies	Y	Y	Y	Y	Y
Other fixed effects and controls	Y	Y	N	Y	Y
Specification				Conditional on trading	Conditional on trading
R^2	0.031	0.380	0.364	0.483	0.154
Observations	22,439	22,439	23,441	6,606	6,606

Notes: Table shows results from regression (3). The unit of observation is a window between two consecutive survey responses by the same individual. The dependent variable in columns 1 and 5 is the change in the equity share due to active trading between the two survey waves. The dependent variable in columns 2 and 3 is an indicator for whether there was any active trading between the two survey waves, defined as an active change in the equity share of at least one percentage point. The dependent variable in column 4 is an indicator of whether the individual actively increased her portfolio share in equity by at least one percentage point during the window as a result of trading between the two survey waves. All columns control for the length of time between two consecutive answers, and for dummies capturing extreme start-of-period equity shares of 0 percent or 100 percent. All columns, except column 3, also control for the respondents' age, gender, region of residence, wealth, and the survey wave. Columns 4

- Figure 3 shows that average beliefs move over time, especially around COVID-19, but cross-sectional dispersion is much larger.
- Table 6 decomposes panel variation: for expected 1-year stock returns, time fixed effects explain 5.0%, individual fixed effects explain 47.5%, and both explain 51.5%.
- Similar individual-fixed-effect dominance appears across expected returns, GDP growth, disaster probabilities, and confidence measures.
- Table 7 shows this result is robust to requiring respondents to answer more survey waves.
- Table 8 shows that demographics and experiences explain only a small fraction of the persistent fixed effects, usually below 7%.

Economic message: Models should not focus only on the time series of average beliefs. Persistent disagreement is the larger empirical object.

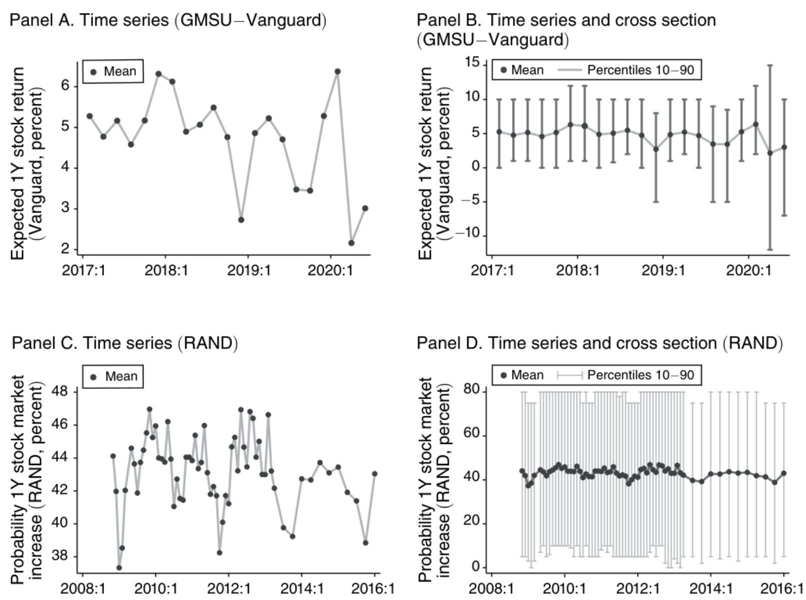

FIGURE 3. TIME-SERIES AND CROSS-SECTIONAL VARIATION: GMSU-VANGUARD AND RAND

Figure 3: time-series and cross-sectional variation

TABLE 6—DECOMPOSING THE VARIATION IN BELIEFS: INDIVIDUAL AND TIME FIXED EFFECTS

	R^2 (percent) of panel regression			
	Time FE	Individual FE	Time + individual FE	Observations
Expected 1Y stock return (percent)	5.0	47.5	51.5	1,960
Expected 10Y stock return (percent p.a.)	0.5	45.0	45.5	1,964
Probability 1Y stock return $< -10\%$	2.7	51.5	53.6	2,011
SD expected 1Y stock return (percent)	0.5	56.7	57.2	2,011
Confidence (stock Qs)	1.4	60.6	62.0	1,988
Expected 3Y GDP growth (percent p.a.)	3.8	43.9	46.8	1,968
Expected 10Y GDP growth (percent p.a.)	0.6	39.7	40.2	1,952
Probability p.a. 3Y GDP growth $< 0\%$	5.1	45.4	49.3	2,010
SD expected p.a. 3Y GDP growth (percent)	1.0	56.5	57.3	2,010
Confidence (GDP Qs)	0.8	62.8	63.8	1,978
Expected 1Y return of 10Y bond (percent)	2.4	38.8	40.7	1,953
Confidence (bond Qs)	0.4	62.9	63.3	1,969

Table 6: individual and time fixed effects

TABLE 7—DECOMPOSING THE VARIATION IN BELIEFS: ROBUSTNESS

	R^2 (total, percent)				Number of individuals			
	#Resp≥5	#Resp≥6	#Resp≥7	#Resp≥8	#Resp≥5	#Resp≥6	#Resp≥7	#Resp≥8
Expected 1Y stock return (percent)	47.5	46.5	46.6	45.9	1,960	1,361	974	712
Expected 10Y stock return (percent p.a.)	45.0	43.7	45.8	44.9	1,964	1,360	959	704
Probability 1Y stock return $< -10\%$	51.5	51.3	52.1	52.3	2,011	1,389	1,003	729
SD expected 1Y stock return (percent)	56.7	57.6	57.7	57.8	2,011	1,389	1,003	729
Confidence (stock Qs)	60.6	60.4	60.9	60.5	1,988	1,374	975	718
Expected 3Y GDP growth (percent p.a.)	43.9	43.2	42.8	39.6	1,968	1,371	978	715
Expected 10Y GDP growth (percent p.a.)	39.7	39.6	38.5	36.7	1,952	1,342	963	708
Probability p.a. 3Y GDP growth $< 0\%$	45.4	44.1	43.9	44.4	2,010	1,392	1,000	730
SD expected p.a. 3Y GDP growth (percent)	56.5	57.5	57.5	58.3	2,010	1,392	1,000	730
Confidence (GDP Qs)	62.8	62.7	62.2	62.1	1,978	1,364	978	721
Expected 1Y return of 10Y bond (percent)	38.8	37.3	35.6	34.7	1,953	1,342	968	705
Confidence (bond Qs)	62.9	62.9	62.6	63.1	1,969	1,363	981	703

Notes: The left panel reports the R^2 values corresponding to regression (5). The right panel reports the number of

Table 7: robustness of belief fixed effects

TABLE 8—BELIEFS HETEROGENEITY AND DEMOGRAPHICS

R^2	#Resp $\geq$ 1	#Resp $\geq$ 2	#Resp $\geq$ 3	#Resp $\geq$ 4	#Resp $\geq$ 5
Expected 1Y stock return (percent)	2.5	4.3	5.3	6.1	6.8
Expected 10Y stock return (percent p.a.)	1.9	3.5	4.4	4.8	5.6
Probability 1Y stock return $< -10\%$	3.0	4.4	6.3	6.5	6.9
SD expected 1Y stock return (percent)	7.1	8.3	10.8	10.2	9.4
Expected 3Y GDP growth (percent p.a.)	2.3	3.0	3.5	4.5	5.3
Expected 10Y GDP growth (percent p.a.)	3.0	3.7	3.8	4.0	4.4
Probability p.a. 3Y GDP growth $< 0\%$	5.0	6.4	8.5	8.5	7.9
SD expected p.a. 3Y GDP growth (percent)	4.8	4.8	5.3	5.1	4.3
Expected 1Y return of 10Y bond (percent)	2.2	3.0	3.0	3.7	4.1

Notes: The table reports the R^2 statistics corresponding to regression (7). In each column, going from left to right, we increase from 1 to 5 the minimum number of responses for an individual to be included in the sample. Each row

Table 8: demographics explain little belief heterogeneity

5.7 Figure 4 and Table 9: expected returns and expected cash-flow growth

Read:

- Figure 4 shows positive relationships between short-run and long-run stock-return expectations and between short-run and long-run GDP-growth expectations.
- It also shows that investors who expect higher GDP growth tend to expect higher stock returns.
- Table 9 confirms the positive relationship in regressions.
- The relationship holds both across investors and within investors after individual fixed effects.

Economic message: Expected returns and expected cash-flow growth are not independent belief objects. Their positive association is an important calibration target for macro-finance models.

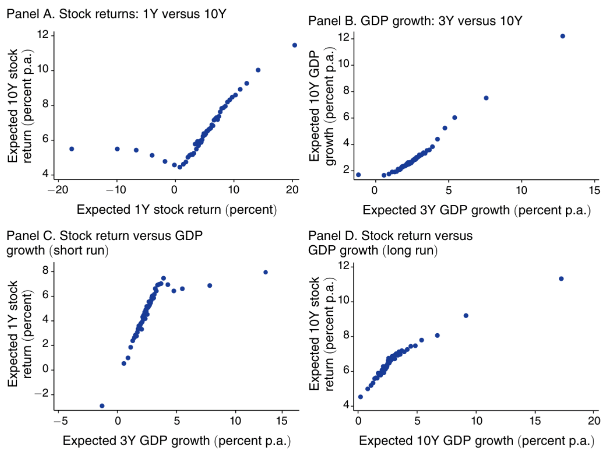

Figure 4. Relationships among Different Beliefs within the Same Survey

Figure 4: relationships among beliefs

Table 9—Correlation across Survey Responses

	Expected 10Y stock returns (percent p.a.)		Expected 10Y GDP growth (percent p.a.)	
	(1)	(2)	(3)	(4)
Panel A				
Expected 1Y stock return (percent)	0.198	0.100		
	(0.006)	(0.009)		
Expected 3Y GDP growth (percent p.a.)			0.824	0.640
			(0.016)	(0.039)
Controls	Y	Y	Y	Y
Individual fixed effects	N	Y	N	Y
R^2	0.111	0.711	0.442	0.810
Observations	42,978	42,978	42,751	42,751
Panel B				
Expected 3Y GDP growth (percent p.a.)	0.710	0.381		
	(0.023)	(0.044)		
Expected 10Y GDP growth (percent p.a.)			0.388	0.260
			(0.015)	(0.035)
Controls	Y	Y	Y	Y
Individual fixed effects	N	Y	N	Y
R^2	0.105	0.712	0.089	0.717
Observations	42,926	42,926	42,543	42,543

Notes: Table shows results from regressing answers to different expectation questions onto each other; panel A

Table 9: correlations across survey responses

5.8 Figure 5 and Table 10: rare disasters and expected returns

Read:

- Figure 5 shows that stock-market disaster probabilities and GDP disaster probabilities are positively related.
- It also shows that investors who assign higher probability to a stock-market disaster expect lower 1-year stock returns.
- Table 10 quantifies the relationship: the coefficient on $Pr(R_{1y} < -30\%)$ is about -0.21 in the baseline cross-sectional specification.
- A 5 percentage point higher subjective stock-disaster probability is associated with about a 1 percentage point lower expected 1-year stock return.
- The negative relationship remains with individual fixed effects, so it also holds within investors over time.

Economic message: This pattern supports heterogeneous rare-disaster beliefs with disagreement: pessimists expect both lower returns and higher disaster probabilities.

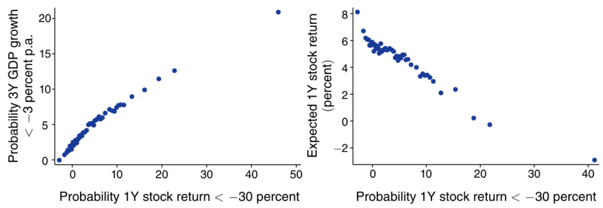

FIGURE 5. STOCK DISASTERS, GDP DISASTERS, AND EXPECTED RETURNS

Notes: The left panel shows a conditional binned scatter plot of survey respondents' subjective probabilities that the 1-year stock returns are below −30 percent and their expectations that annualized average GDP growth over the next three years is below −3 percent. The right panel shows a conditional binned scatter plot of survey respondents' subjective probabilities that the 1-year stock returns are below −30 percent and their expected 1-year stock returns.

Figure 5: disasters and expected returns

TABLE 10—EXPECTED STOCK RETURNS AND RARE DISASTERS BELIEFS

	Expected 1Y stock return (percent)					
	(1)	(2)	(3)	(4)	(5)	(6)
Probability 1Y stock return < −30%	−0.212 (0.008)		−0.270 (0.013)		−0.146 (0.013)	−0.153 (0.017)
Probability 1Y stock return < −10%		−0.152 (0.003)				
Probability 1Y stock return < −30% × low bucket first				−0.208 (0.010)		
Probability 1Y stock return < −30% × high bucket first				−0.217 (0.012)		
Controls	Y	Y	Y	Y	Y	Y
Individual fixed effects					Y	Y
Specification			Prob. ∈ [0.1%, 10%]			Prob. ∈ [0.1%, 10%]
R^2	0.117	0.244	0.055	0.438	0.722	0.764
Observations	43,492	43,492	23,466	43,492	43,492	22,670

Notes: Table shows results from regression (8). The unit of observation is a survey response; the dependent variable is the expected 1-year stock return. Columns 1 to 4 control for the respondents' age, gender, region of residence, wealth, and the survey wave. Columns 5 and 6 include individual fixed effects, survey wave fixed effects, and a dummy for the randomization order of the buckets in the distribution question. Columns 3 and 6 restrict the sample to individuals who report expected probabilities of a stock market disaster between 0.1 percent and 10 percent. Standard errors are clustered at the respondent level

Table 10: rare-disaster beliefs and expected returns

6 Short summary

- The paper creates and analyzes a new survey of wealthy Vanguard investors.
- The survey elicits quantitative beliefs about stock returns, GDP growth, bond returns, subjective distributions, confidence, and difficulty.
- The key strength is that survey responses are matched to real administrative portfolio, trading, and login data.
- Beliefs are related to portfolios: higher expected stock returns imply higher equity shares.
- The average sensitivity is small: about 0.7 percentage points of equity share for a 1 percentage point higher expected stock return.
- This low sensitivity is not mainly explained by classical measurement error.
- Heterogeneity matters: more attentive, active, confident, wealthy, and tax-advantaged investors have higher pass-through.
- Belief changes do not strongly predict whether an investor trades.
- Conditional on trading, belief changes affect whether the investor buys equity and how much the equity share changes.
- Most belief variation is persistent cross-sectional heterogeneity, not common time-series movement.
- Observable demographics explain little of this persistent heterogeneity.
- Investors who expect higher GDP growth also expect higher stock returns.
- Investors who perceive stock-market disasters as more likely expect lower stock returns.
- The paper's five facts are meant as discipline for macro-finance models that use survey expectations.

7 Conclusion

7.1 Core conclusion

Giglio, Maggiori, Stroebel, and Utkus show that survey beliefs matter for real portfolio behavior, but not in the frictionless representative-agent way. Beliefs pass through to portfolios positively, yet weakly on average. The large empirical objects are heterogeneity, persistence, infrequent trading, and differences in investor attention and confidence.

7.2 Economic intuition

The intuition is that investors do not continuously rebalance Merton portfolios. They hold sticky allocations, trade infrequently, face tax and account frictions, pay varying levels of attention, and differ in confidence. Once these frictions are recognized, it becomes reasonable that survey beliefs are informative but do not mechanically translate one-for-one into portfolio shares.

7.3 Takeaway

Survey-based macro-finance evidence is useful and should be taken seriously. But the models that use it should feature heterogeneous beliefs, persistent optimism and pessimism, infrequent trading, and weak average pass-through from beliefs to demand. The paper's five facts provide direct empirical targets for those models.